



QpiAI™ - Joint AI Training and Certification Program

Feature Engineering, Hyperparameter Tuning

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Recap:

**Basic Probability &
Statistics**

**Regression and
Classification**

**Linear & Logistic
Regression**

**K-Nearest Neighbor
Algorithm**

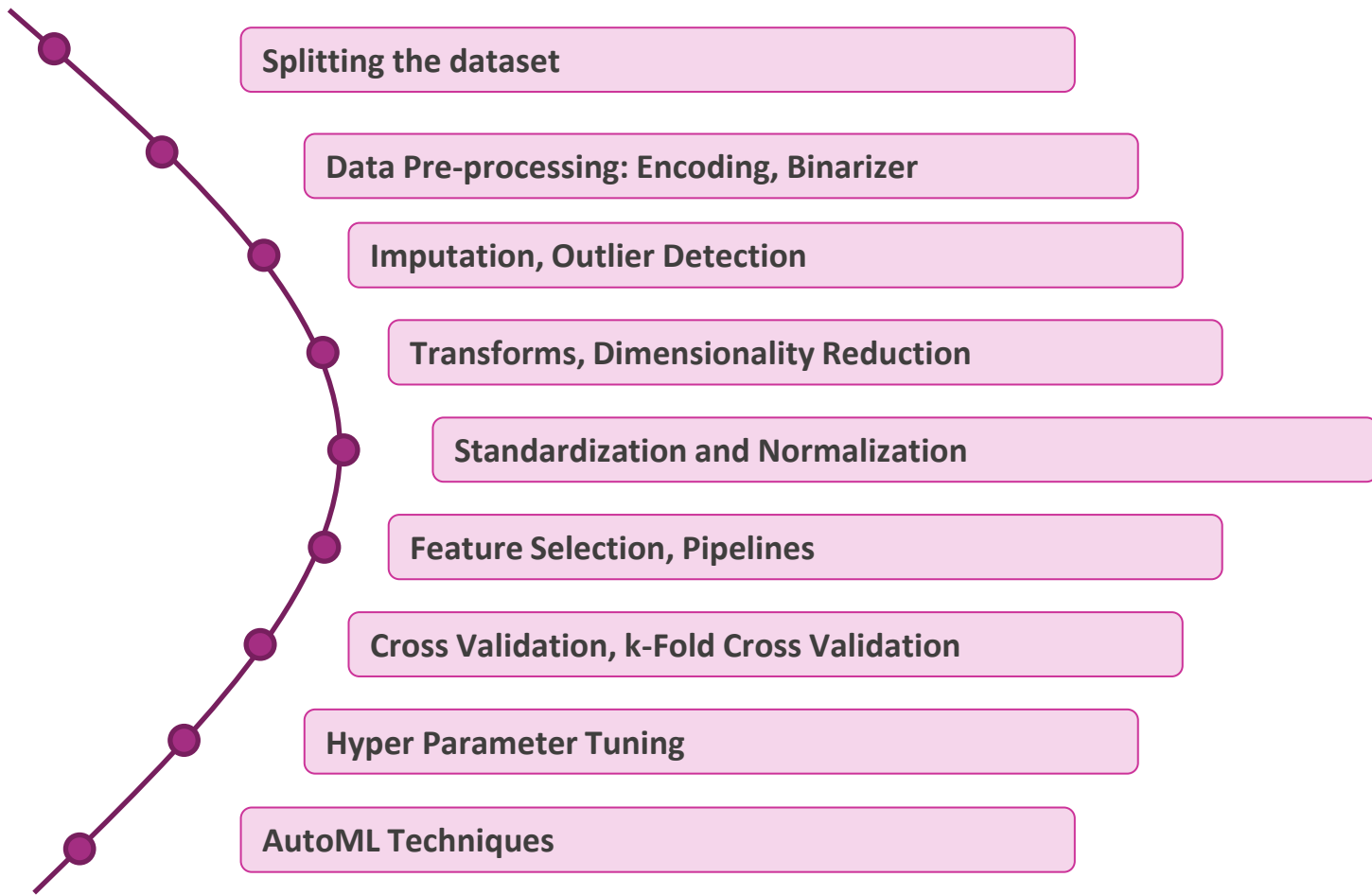
**Support Vector
Machines, Classification
Performance Metrics**

**Decision Tree,
Random Forest &
Ensemble Methods**

**Principal Component
Analysis and k- Means**

Time Series Analysis

Overview



Nominal

A variable with categories that does not have a natural order or ranking. Can code nominal variables with numbers, but the order is arbitrary and any calculations, such as computing a mean, median, or standard deviation, is meaningless. Eg: blood group, zip code, ethnicity, eye color.

Ordinal

An ordinal scale is one where the order matters but not the difference between values. Eg:

- Socio economic status: low income, middle income , high income
- Education level: high school, undergraduate, graduate, doctoral

Interval

A variable following interval scale is one where there is order and the difference between two values is meaningful.

Eg: Temperature in Celsius, pH, Credit Score.

Ratio

A ratio variable, has all the properties of an interval variable, and also has a clear definition of 0. When the variable equals 0, there is none of that variable. Eg: Weight, Temperature in Kelvin.

Dataset Encoding

- Splitting the dataset doesn't always ensure whether it is suitable for training with the model.
- Models are mathematical tools and expects only numerical values in the dataset.
- But datasets doesn't always have numerical values only. It has all kind of categorical, string etc. valued columns. So we need to perform various kinds of dataset encoding so that it is in readable format by the model.
- There are various kinds of Dataset encodings we can perform like label encoding, one-hot encoding, label binarizer, etc.
- Label Binarizer: Binarizes data (set feature values to 0 or 1) according to a threshold (default 0), values greater than the threshold map to 1, while values less than or equal to the threshold map to 0. With the default threshold of 0, only positive values map to 1.

One Hot Encoding

It refers to splitting the column which contains numerical categorical data to many columns depending on the number of categories present in that column. Each column contains “0” or “1” corresponding to which column it has been placed.

Foundation	Type_is_BrkTil	Type_is_CBlock	Type_is_PConc	Type_is_Slab	Type_is_Stone	Type_is_Wood
PConc	0	0	1	0	0	0
CBlock	0	1	0	0	0	0
PConc	0	0	1	0	0	0
BrkTil	1	0	0	0	0	0
PConc	0	0	1	0	0	0
Wood	0	0	0	0	0	1
PConc	0	0	1	0	0	0
CBlock	0	1	0	0	0	0
BrkTil	1	0	0	0	0	0

Label Encoding

Label Encoder encodes labels within a categorical column with a value between 0 and $n_classes - 1$ where $n_classes$ is the number of distinct labels.

Label encoding can work on Numeric data too, but it is preferable to use with Categorical data.

Foundation	Foundation_label_encoded
PConc	2
CBlock	1
PConc	2
BrkTil	0
PConc	2
Wood	5
PConc	2
CBlock	1
BrkTil	0

